



# Wake-on-LAN

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**Wake-on-LAN** [↗](#) (WoL) is a feature to switch on a computer via the network.

## 1 Hardware settings

Wake-On-LAN only works if you fulfil the following preconditions:

1. The target computer's motherboard and [Network Interface Controller](#) [↗](#) has to support Wake-on-LAN.
2. The target computer has to be *physically* connected (with a cable) to a router or to the source computer for WoL to work properly *unless* your wireless card has support for Wake on Wireless (WoWLAN or WoW).

You have to prepare the following in your BIOS/UEFI:

1. Enable the Wake-on-LAN feature. Different motherboard manufacturers use slightly different language for this feature. Look for terminology such as "PCI Power up", "Allow PCI wake up event" or "Boot from PCI/PCI-E".
2. (If available in BIOS/UEFI) Make sure that [ErP](#) [↗](#) is **disabled**, otherwise your Ethernet card will not be powered and will not be able to receive any wake-up packets sent from another device.

### Note:

- Some motherboards support Wake-on-LAN from a powered-off state, but some only support Wake-on-LAN from a sleep / suspended state.
- Some motherboards are affected by a bug that can cause immediate or random [#Wake-up after shutdown](#) whenever the BIOS WoL feature is enabled.

## 2 Software configuration

### 2.1 Enable WoL on the network adapter

Depending on the hardware, the network driver may have WoL switched off by default.

To query this status or to change the settings, install [ethtool](#), determine the name of the [network interface](#), and

query it using the command:

```
# ethtool interface | grep Wake-on
```

```
Supports Wake-on: pumbag
Wake-on: d
```

The *Wake-on* values define what activity triggers wake up: **d** (disabled), **p** (PHY activity), **u** (unicast activity), **m** (multicast activity), **b** (broadcast activity), **a** (ARP activity), and **g** (magic packet activity). The value **g** is required for WoL to work, if not, the following command enables the WoL feature in the driver:

```
# ethtool -s interface wol g
```

**Note:** Setting one of **u**, **m** or **b** along with **g** might also be necessary to enable the feature.

This command might not last beyond the next reboot and in this case must be repeated via some mechanism. Common solutions are listed in the following subsections.

## 2.2 Make it persistent

### 2.2.1 systemd.link

Link-level configuration is possible through [systemd.link files](#). The actual setup is performed by the `net_setup_link` udev builtin. Add the `WakeOnLan` option to the network link file:

```
/etc/systemd/network/50-wired.link
```

```
[Match]
MACAddress=aa:bb:cc:dd:ee:ff

[Link]
NamePolicy=kernel database onboard slot path
MACAddressPolicy=persistent
WakeOnLan=magic
```

Also see [systemd.link\(5\)](#) for more information.

**Note:**

- Only the first matching file is applied. The content of the default link file `/usr/lib/systemd/network/99-default.link` shipped with systemd has to be included, otherwise the interface might be misconfigured.
- To be considered, the file name should alphabetically come before the default `99-default.link`. For example `50-wired.link` works.
- This configuration applies only to the link-level, and is independent of network-level daemons such as [NetworkManager](#) or [systemd-networkd](#).
- In the `Match` section, `OriginalName=` can also be used to identify the interface.

### 2.2.2 systemd service

This is an equivalent of previous `systemd.link` option, but uses a standalone systemd service.

```
/etc/systemd/system/wol@.service

[Unit]
Description=Wake-on-LAN for %i
Requires=network.target
After=network.target

[Service]
ExecStart=/usr/bin/ethtool -s %i wol g
Type=oneshot

[Install]
WantedBy=multi-user.target
```

Alternatively install the `wol-systemd`<sup>AUR</sup> package, then activate this new service by **starting** `wol@interface.service`.

### 2.2.3 udev

`udev` is capable of running any command as soon as a device is visible. The following rule will turn on WOL on all **network interfaces** whose name matches `en*`. The file name is important and must start with a number between 81 and 99 so that it runs **after** `80-net-setup-link.rules`, which renames interfaces with predictable names. Otherwise, `NAME` would be undefined and the rule would not run.

```
/etc/udev/rules.d/81-wol.rules

ACTION=="add", SUBSYSTEM=="net", NAME=="en*", RUN+="/usr/bin/ethtool -s $name wol g"
```

The `$name` placeholder will be replaced by the value of the `NAME` variable for the matched device.

### 2.2.4 cron

A command can be run each time the computer is (re)booted using `"@reboot"` in a crontab. First, make sure **cron** is enabled, and then **edit a crontab** for the root user that contains the following line:

```
@reboot /usr/bin/ethtool -s interface wol g
```

### 2.2.5 netctl

If using **netctl**, one can make this setting persistent by adding the following the netctl profile:

```
/etc/netctl/profile

ExecUpPost='/usr/bin/ethtool -s interface wol g'
```

### 2.2.6 NetworkManager

NetworkManager provides **Wake-on-LAN ethernet support**<sup>🔗</sup>. One way to enable Wake-on-LAN by magic

packet is through *nmcli*.

First, search for the name of the wired connection:

```
# nmcli con show
```

| NAME   | UUID                                 | TYPE           | DEVICE  |
|--------|--------------------------------------|----------------|---------|
| wired1 | 612e300a-c047-4adb-91e2-12ea7bfe214e | 802-3-ethernet | enp0s25 |

By following, one can view current status of Wake-on-LAN settings:

```
# nmcli c show "wired1" | grep 802-3-ethernet.wake-on-lan
```

```
802-3-ethernet.wake-on-lan:      default
802-3-ethernet.wake-on-lan-password:  --
```

Enable Wake-on-LAN by magic packet on that connection:

```
# nmcli c modify "wired1" 802-3-ethernet.wake-on-lan magic
```

Then reboot, possibly two times. To disable Wake-on-LAN, substitute `magic` with `ignore`.

The Wake-on-LAN settings can also be changed from the GUI using [nm-connection-editor](#).

You can disable Wake-on-LAN for all connections permanently by adding a dedicated configuration file :

```
/etc/NetworkManager/conf.d/wake-on-Lan.conf
```

```
[connection]
ethernet.wake-on-lan = ignore
wifi.wake-on-wlan = ignore
```

## 2.3 Enable WoL in TLP

When using [TLP](#) for suspend/hibernate, the `WOL_DISABLE` setting should be set to `N` in `/etc/tlp.conf` to allow resuming the computer with WoL.

## 3 Trigger a wake up

To trigger WoL on a target machine, its **MAC address** must be known. To obtain it, execute the following command from the machine:

```
$ ip link
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp1s0: <BROADCAST,MULTICAST,PROMISC,UP,LOWER_UP> mtu 1500 qdisc fq_codel master br0
   state UP group default qlen 1000
   link/ether 48:05:ca:09:0e:6a brd ff:ff:ff:ff:ff:ff
```

Here the MAC address is `48:05:ca:09:0e:6a`.

In its simplest form, Wake-on-LAN broadcasts the magic packet as an ethernet frame, containing the MAC address within the current network subnet, below the IP protocol layer. The knowledge of an IP address for the target computer is not necessary, as it operates on layer 2 (Data Link).

If used to wake up a computer over the internet or in a different subnet, it typically relies on the router to relay the packet and broadcast it. In this scenario, the external IP address of the router must be known. Keep in mind that most routers by default will not relay subnet directed broadcasts as a safety precaution and need to be explicitly told to do so.

Applications that are able to send magic packets for Wake-on-LAN:

- **gWakeOnLAN** — GTK utility to awake turned off computers through the Wake-on-LAN feature.  
<https://www.muflone.com/gwakeonlan/english/> || **gwakeonlan**
- **wol** — Implements Wake-on-LAN functionality in a small program. It wakes up hardware that is Magic Packet compliant. Note: This application will need the port changed to 9 from the default(40000) using the -p argument/flag.  
<https://sourceforge.net/projects/wake-on-lan/> || **wol**
- **wol\_qt** — Qt utility with integrated ARP scan to find MAC addresses and batch sending of Wake-on-LAN packets.  
[https://github.com/stefmitropoulos/wol\\_qt](https://github.com/stefmitropoulos/wol_qt) || **wol\_qt**<sup>AUR</sup>

### 3.1 On the same LAN

If you are connected directly to another computer through a network cable, or the traffic within a LAN is not firewalled, then using Wake-on-LAN should be straightforward since there is no need to worry about port redirects.

In the simplest case the default broadcast address `255.255.255.255` is used:

```
$ wol target_MAC_address
```

To broadcast the magic packet only to a specific subnet or host, use the `-i` switch:

```
$ wol -i target_IP target_MAC_address
```

### 3.2 Across the internet

When the source and target computers are separated by a NAT router, different solution can be envisaged:

- If the router supports *WoL*, one can rely on it to properly broadcast the packet into the local network.

Otherwise Wake-on-LAN can be achieved via [port forwarding](#). The router needs to be configured using one of these two options:

- Forward a different port to each target machine. This requires any target machine to have a static IP address on its LAN.
- Forward a single port to the [broadcast address](#). Most routers do not allow to forward to broadcast, however if you can get shell access to your router, through telnet, ssh, serial cable or other mean, run the command:

```
$ ip neighbor add 192.168.1.254 lladdr FF:FF:FF:FF:FF:FF dev net0
```

This example assumes the network is *192.168.1.0/24* and uses *net0* as network interface. Now, forward UDP port 9 to 192.168.1.254. This solution was successfully tested on a Linksys WRT54G running [Tomato](#), and on the Verizon FIOS ActionTec router. For notes on how to do it on a router with [DD-WRT](#) firmware, see [this tutorial](#) and for a router with [OpenWrt](#) firmware, see [this tutorial](#).

In any case, run the following command from the source computer to trigger wake-up:

```
$ wol -p forwarded_port -i router_IP target_MAC_address
```

## 4 Miscellaneous

### 4.1 Check reception of the magic packets

In order to make sure the WoL packets reach the target computer, one can listen to the UDP port, usually port 9, for magic packets. The magic packet frame expected contains 6 bytes of FF followed by 16 repetitions of the target computer's MAC (6 bytes each) for a total of 102 bytes.

#### 4.1.1 Using netcat

This can be performed by installing [gnu-netcat](#) on the target computer and using the following command:

```
# nc --udp --listen --local-port=9 --hexdump
```

Then wait for the incoming traffic to appear in the `nc` terminal.

**Note:** The firewall on the machine itself does not need to be opened for wake-on-LAN to work (interface's processing happen in the NIC, before the firewall). However, for debugging purposes with netcat, you still need to temporarily open that port.

#### 4.1.2 Using ngrep

Install [ngrep](#) on the target computer and type the following command:

```
# ngrep '\xff{6}(.{6})\1{15}' -x port 9
```

## 4.2 Example of WoL script

Here is a script that illustrates the use of `wol` with different machines:

```
#!/bin/bash

# definition of MAC addresses
monster=01:12:46:82:ab:4f
ghost=01:1a:d2:56:6b:e6

echo "Which PC to wake?"
echo "m) monster"
echo "g) ghost"
```

```

echo "q) quit"
read input1
case $input1 in
    m)
        /usr/bin/wol $monster
        ;;
    g)
        # uses wol over the internet provided that port 9 is forwarded to ghost on ghost's
        router
        /usr/bin/wol --port=9 --host=ghost.mydomain.org $ghost
        ;;
    Q|q)
        break
        ;;
esac

```

## 5 Troubleshooting

### 5.1 NetworkManager

#### 5.1.1 Network adapter is still powered off on shutdown

Setting auto negotiation to `yes` may help if WOL is configured through *nmcli* and network adapter is still powered off on shutdown.

Set it using:

```
# nmcli c modify "wired1" 802-3-ethernet.auto-negotiate yes
```

### 5.2 Wake-up after shutdown

It is known that some motherboards are affected by a bug that can cause immediate or random wake-up after a *shutdown* whenever the BIOS WoL feature is enabled (as discussed in [this thread](#) for example).

#### 5.2.1 Fix using BIOS Settings

The following actions in the BIOS preferences can solve this issue with some motherboards:

1. Disable all references to *xHCI* in the USB settings (note this will also disable USB 3.0 at boot time)
2. Disable *EuP 2013* if it is explicitly an option
3. Optionally enable wake-up on keyboard actions

**Note:** There are mixed opinions as to the value of #3 above and it may be motherboard dependent.

#### 5.2.2 Fix by kernel quirks

The issue can also be solved by adding the following kernel boot parameter: `xhci_hcd.quirks=270336` This activates the following quirks:

- `XHCI_SPURIOUS_REBOOT`
- `XHCI_SPURIOUS_WAKEUP`

## 5.3 Battery draining problem

Some laptops have a battery draining problem after shutdown [\[1\]](#). This might be caused by enabled WOL. To solve this problem, disable it by using `ethtool` as mentioned above.

```
# ethtool -s net0 wol d
```

## 5.4 Realtek

### 5.4.1 r8168

Users with a Realtek 8168 8169 8101 8111(C) based NICs (cards / and on-board) may notice a problem where the NIC seems to be disabled on boot and has no link light. See [Network configuration/Ethernet#Realtek no link / WOL problem](#).

If the link light on the network switch is enabled when the computer is turned off but Wake on LAN is still not working, booting the system using the `r8168`<sup>AUR</sup> kernel module at least once and then switching back to the `r8169` kernel module included with the kernel has been reported to fix it.

For the `r8168` module you might need to set the `s5wol=1` [kernel module parameter](#) to enable the wake on LAN functionality.

### 5.4.2 r8125

Users with a Realtek 8125 NIC have reported being unable to use the Wake on LAN feature with the `r8169` kernel module. Installing `r8125-dkms`<sup>AUR</sup> enables the functionality.

## 5.5 alx driver support

For some newer Atheros-based NICs (such as Atheros AR8161 and Killer E2500), WOL support has been disabled in the mainline `alx` module due to a bug causing unintentional wake-up (see [this patch discussion](#)). A patch can be applied (or installed as a `dkms` module using the `alx-wol-dkms`<sup>AUR</sup> package) which both restores WOL support and fixes the underlying bug, as outlined in [this thread](#).

See also the pre-patched sources in [\[2\]](#).

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